Project Proposal

Project: MONTIG | مُنْتِج: Social Media Impact Analysis & Digital Wellbeing Ecosystem.

Objective: The primary objective of **MONTIG | مُنْتِج** is to design and develop an enterprise-grade **Power BI Business Intelligence platform** that analyzes the impact of social media consumption on individual productivity, sleep quality, and lifestyle patterns. The core of the project relies heavily on advanced **Power BI data modeling, complex DAX measures, dynamic KPIs, and interactive reporting dashboards**. To maximize project value, supplementary features including an advanced **SQL query layer**, a **Machine Learning predictive model**, and a **Digital Wellbeing Guidance Website** are integrated as value-added components.

Description: The core deliverable of **MONTIG | مُنْتِج** is an advanced **Power BI analytical suite** featuring **5 specialized interactive dashboards**, **20 dynamic KPIs**, and complex **DAX Measures** built upon a star-schema dataset of 29,869 records and 19 features. While Power BI serves as the main decision-support hub, the project is enhanced with supplementary layers including **29 SQL business queries**, a **predictive Machine Learning feature**, and a **Digital Wellbeing Web Guide**.

Tools & Design:

- **Primary Business Intelligence & Analytics:** Power BI Desktop, Data Analysis Expressions (**DAX**), Star Schema Data Modeling, Custom Calculated Columns & Measures, Interactive Slicers, Drill-throughs, and Custom Tooltips.
- **ETL & Data Preprocessing:** Power Query (M Language) for data transformation, cleaning, and shaping.
  - **Supplementary Add-on Stack:**
  - SQL Server (Back-end data querying and validation)
  - Python (scikit-learn, Pandas) for Machine Learning predictions
  - HTML/CSS/JavaScript / Streamlit for the Web Guidance Portal

Technologies to Use:

- Primary: Power BI Desktop & Power BI Service
- Add-ons: SQL Server / Python / ML / Web

Team Members & Responsibilities:

1- Ahmed Fathallah Saad Ahmed

- **Role:** Lead Data Collector, Machine Learning Specialist & Project Maintenance Lead
- **Responsibilities:** Spearheading data collection, project ideation, developing and training the Machine Learning predictive model, and overseeing comprehensive system maintenance across Power BI dashboards, SQL database, and the web platform.

2- Mohamed Mohamed Ahmed Gaballah

- **Role:** Business Domain & SQL Questions & **Dashboard 01** Lead Developer
- **Responsibilities:** Designing database schemas, writing advanced SQL queries, and building **Dashboard 01: Social Media Usage Overview** in Power BI.

3- Samaa Elsayed Mohammed AbdEllah Sliem

- **Role:** Data Cleaning Specialist & **Dashboard 02** Lead Developer
- **Responsibilities:** Executing data cleaning and preprocessing pipelines, developing and styling the supplementary Digital Wellbeing Website, and building **Dashboard 02: Productivity & Work Performance** in Power BI.

4- Sama Hatem Abdellatife El Sayed Ahmed

- **Role:** Power BI Developer & **Dashboard 03** Lead Developer
- **Responsibilities:** Writing complex DAX measures, performing statistical analysis, and building **Dashboard 03: Wellbeing, Sleep & Burnout** in Power BI.

5- Habiba Alaa Ebrahim Abdelrasol

- **Role:** Power BI Developer & **Dashboard 04** Lead Developer
- **Responsibilities:** Power Query ETL transformation, optimizing the data model, and building **Dashboard 04: Digital Habits & Focus** in Power BI.

6- Farah Ahmed Mohamed Abdelrady

- **Role:** Business Analyst, Documentation Lead & **Dashboard 05** Lead Developer
- **Responsibilities:** Requirements gathering, drafting business questions, technical report documentation, managing final presentations, and building **Dashboard 05: Workforce Segmentation & Insights** in Power BI.

- **Week 1: Data Collection, Idea Formulation & Power Query ETL**
    - *Tasks:* Executing initial data collection and project ideation (Ahmed Fathallah); performing data cleaning, handling missing values, and shaping datasets via Power Query (Samaa Elsayed); setting up SQL database schemas (Mohamed Mohamed); defining core business questions, KPI frameworks, and DAX requirements (Farah Ahmed & Team).
    - *Deliverables:* Cleaned, relational dataset ready for Power BI ingestion and initialized SQL database structure.

- **Week 2: Advanced DAX Architecture, SQL Analytics & ML Modeling**
    - *Tasks:* Writing and testing 29 complex SQL analytical queries (Mohamed Mohamed); developing and training the Machine Learning predictive model (Ahmed Fathallah); designing Power BI wireframes and implementing foundational DAX measures (All Dashboard Leads).
    - *Deliverables:* Validated SQL query repository, trained Machine Learning prediction engine, and structural Power BI data model.

- **Week 3: Power BI Dashboard Implementation & Web Development**
- *Tasks:* Constructing and formatting the 5 core Power BI analytical dashboards with rich DAX measures and interactive slicers:
    - *Dashboard 01:* Social Media Usage Overview (Mohamed Mohamed)
    - *Dashboard 02:* Productivity & Work Performance (Samaa Elsayed)
    - *Dashboard 03:* Wellbeing, Sleep & Burnout (Sama Hatem)
    - *Dashboard 04:* Digital Habits & Focus (Habiba Alaa)
    - *Dashboard 05:* Workforce Segmentation & Insights (Farah Ahmed)
- *Tasks:* Developing and deploying the supplementary Digital Wellbeing Guidance Website (Samaa Elsayed).

- *Deliverables:* Fully interactive 5-page Power BI Report file (.pbix) and operational web portal.

- **Week 4: BI Optimization, Quality Assurance, Documentation & Presentation**
    - *Tasks:* Conducting end-to-end testing, performance analyzer optimization for DAX measures, and cross-system maintenance (Ahmed Fathallah & All Team); compiling technical documentation and pushing repositories to GitHub (Farah Ahmed); preparing executive presentation decks and the DEPI Infographic summary for **MONTIG |** مُنْتِج(Farah Ahmed & All Team).
    - *Deliverables:* Fully optimized Power BI report, published GitHub repository, and final presentation deck.

Project Governance & Requirements

1. Comprehensive Risk Assessment & Mitigation Framework (Power BI)

- **Data Integrity and Model Anomalies:** Missing or inconsistent records within the 30,000-row dataset could distort DAX metrics and visual aggregations. This is mitigated by implementing robust Power Query (M Language) ETL cleaning pipelines and schema validation rules prior to Power BI ingestion.
- **DAX Computational and Rendering Bottlenecks:** Complex measures, filters, and high-cardinality relationships can degrade Power BI visual performance. This is mitigated through optimized DAX formula design, leveraging variables (VAR) to manage evaluation contexts, avoiding iterative row-level functions, and conducting performance tuning via Power BI Performance Analyzer.
- **Add-On Integration Complexity:** Synchronizing auxiliary components (the SQL query layer, Machine Learning script, and Web Guide) with the primary Power BI reporting suite introduces coordination risks. This is mitigated by strictly prioritizing the 5 core Power BI dashboards as the primary deliverable, treating supplementary components as value-added add-ons.
- **Version Control and Repository Conflicts:** Concurrent development of the Power BI report (.pbix) file across multiple team members can lead to version overlaps. This is mitigated by maintaining structured GitHub repositories, defining clear module ownership, and coordinating merge reviews.

2. Stakeholder Architecture & Functional User Stories

- **Stakeholder Architecture:**
    a. **Executive Leadership & HR Management:** Primary stakeholders who rely on the 5 core Power BI analytical dashboards to track organizational burnout trends, productivity scores, and social media impacts.

    b. **Departmental Leads & Team Supervisors:** Operational users who leverage Power BI interactive drill-throughs, slicers, and dynamic KPIs to assess digital habits and workforce wellbeing.

    c. **Workforce & Individual Contributors:** End-users who interact with the supplementary Digital Wellbeing Guidance Web Portal to self-assess habits, while being evaluated through the primary Power BI suite.

- **Functional User Stories:**
    a. **Executive BI Oversight:** *As an HR Manager*, I require centralized, interactive Power BI dashboards mapping workforce burnout against daily social media consumption, enabling data-driven organizational policies.

b. **Operational Performance Tracking:** *As a Team Lead*, I require drill-through capabilities in Power BI linking pre-sleep screen time to actual productivity scores, allowing me to optimize workflow structures.

c. **Value-Added Web Guidance:** *As an Employee*, I require access to the supplementary Digital Wellbeing Web Guide to receive practical recommendations based on insights derived from our analytical models.

3. Non-Functional System Requirements

- **Power BI Analytical Responsiveness:** All 5 Power BI report pages, interactive slicers, and custom visuals must load and render completely within an average threshold of 3 seconds under standard workstation environments.

- **DAX & Modeling Efficiency:** The star-schema data model and custom DAX measures must maintain high query execution efficiency, ensuring minimal latency during cross-filtering.

- **UI/UX Visual Elegance:** Power BI dashboards must adhere to professional design standards, featuring consistent color palettes, structured visual hierarchies, and customized descriptive tooltips.

- **Scalability and Supplementary Add-On Reliability:** The primary Power BI architecture must remain modular and stable, cleanly integrating backend data validation (SQL) and predictive logic (ML) as supportive add-ons.

4. Key Project Performance Indicators (Project KPIs)

- **Dashboard Rendering Latency:** Average DAX execution time of less than 1.5 seconds per visual element during interactive user filtering across the 5 Power BI report pages.
- **Analytical Completeness:** Successful deployment of 5 interactive Power BI dashboards driven by 20 dynamic KPIs and advanced star-schema modeling.
- **System Stability & Add-On Integration:** Maintaining 99.9% uptime for the supplementary web portal, alongside 100% code integrity for the primary .pbix file, SQL queries, and ML prediction scripts on GitHub.
- **Evaluation Alignment:** Successful demonstration of core business intelligence reporting, stakeholder engagement, and technical execution during final assessments.

MONTIG

Gantt chart · 4 weeks · 20 working days

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Deliverables

- **Week 1** — Cleaned relational dataset ready for Power BI ingestion and initialized SQL database structure
- **Week 2** — Validated SQL query repository, trained ML prediction engine and structural Power BI data model
- **Week 3** — Fully interactive 5-page Power BI report (.pbix) and operational web portal
- **Week 4** — Fully optimized Power BI report, published GitHub repository and final presentation deck